

System of Linear Equations

Linear Equation: An equation in which the power of each unknown is one is called a linear equation. The general form of a linear equation is defined as,

$$a_1x_1 + a_2x_2 + a_3x_3 + \dots + a_nx_n = b \dots \dots \dots (1)$$

where, $a_1, a_2, a_3, \dots, a_n$ and b real numbers and $x_1, x_2, x_3, \dots, x_n$ are unknowns (or variables) which is to be determined.

If $b = 0$ then the equation (1) is called a homogeneous linear equation and if $b \neq 0$ then it is called a non-homogeneous linear equation.

Example: 1. $ax + by = c$ (non-homogeneous) represents a straight line.

2. $ax + by + cz + d = 0$ (homogeneous) represents a plane.

Solutions of Linear Equation: A solution of the linear equation

$$a_1x_1 + a_2x_2 + a_3x_3 + \dots \dots + a_nx_n = b$$

in n variables is a sequence of n numbers, $\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_n$ such that the equation is satisfied when we substitute $x_1 = \alpha_1, x_2 = \alpha_2, \dots, x_n = \alpha_n$.

The set of all such solutions of this linear equation is called a solution set.

System of Linear Equations: A finite set of linear equations is known as a system of linear equations. So,

[illegible]

is a system of m linear equations in n variables $x_1, x_2, x_3, \dots \dots \dots x_n$.

If $b_i = 0$, then the above system is called a homogeneous system of linear equations and if $b_i \neq 0$, then it is called a non-homogeneous system of linear equations.

Classification of System of Linear Equations: Regarding the nature of solutions, systems of linear equations are classified as follows:

1. Inconsistent: A system of linear equations is called an inconsistent if it has no solution.

Augmented matrix: Consider a non-homogeneous system of m linear equations in n unknowns is,

$$\left. \begin{array}{l} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2 \\ \dots\dots\dots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m \end{array} \right\} \dots\dots\dots (1) \text{ where, } a_{ij}, b_i \in R, m \geq 2$$

Where, $A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$, $X = \begin{pmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{pmatrix}$ and $B = \begin{pmatrix} b_1 \\ b_2 \\ \dots \\ b_m \end{pmatrix}$

The matrix $[A \quad B] = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} & \vdots & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & \vdots & b_2 \\ \dots & \dots & \dots & \dots & \vdots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} & \vdots & b_m \end{pmatrix}$ is called augmented matrix of the

NOTE:

- 3

Gauss Elimination Method: Consider a non-homogeneous system of m linear equations in n unknowns is,

[illegible]

we can write it as $AX = B$.

Where, $A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$, $X = \begin{pmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{pmatrix}$ and $B = \begin{pmatrix} b_1 \\ b_2 \\ \dots \\ b_m \end{pmatrix}$

The augmented matrix is,

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} & \vdots & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & \vdots & b_2 \\ \dots & \dots & \dots & \dots & \vdots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} & \vdots & b_m \end{pmatrix}$$

reduce this matrix into the following form,

$$\begin{pmatrix} a'_{11} & a'_{12} & \dots & a'_{1n} & \vdots & l_1 \\ 0 & a'_{22} & \dots & a'_{2n} & \vdots & l_2 \\ \dots & \dots & \dots & \dots & \vdots & \dots \\ 0 & 0 & \dots & a'_{mn} & \vdots & l_m \end{pmatrix}$$

then the reduced system is,

[illegible]

Now, by back substitution, we solve for, $x_n, x_{n-1}, \dots, x_1$.

This process which eliminates unknowns from succeeding equations is known as Gauss elimination.

NOTE:

1. If an equation $0x_1 + 0x_2 + \dots + 0x_n = b$, $b \neq 0$ occurs, then the system is inconsistent and has no solution.

- Gauss Jordan Elimination Method:** Consider a non-homogeneous system of m linear equations in n unknowns is,

we can write it as $AX = B$.

The augmented matrix is,

reduce this matrix into the following form,

then the reduced system is,

Thus, we get $x_1 = l_1, x_2 = l_2, \dots \dots x_n = l_m$.

Problem-01: Show that the following system of linear equations is not consistent

$$\begin{aligned}x + y + z &= -3 \\3x + y - 2z &= -2 \\2x + 4y + 7z &= 7\end{aligned}$$

Solution: The given system of linear equations is,

$$\left. \begin{aligned}x + y + z &= -3 \\3x + y - 2z &= -2 \\2x + 4y + 7z &= 7\end{aligned} \right\} \dots \dots \dots (1)$$

the system (1) can be written as,

$$AX = B \dots \dots \dots (2)$$

$$\text{where, } A = \begin{pmatrix} 1 & 1 & 1 \\ 3 & 1 & -2 \\ 2 & 4 & 7 \end{pmatrix}, X = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \text{ and } B = \begin{pmatrix} -3 \\ -2 \\ 7 \end{pmatrix}.$$

The augmented matrix is,

$$\begin{aligned}[A, B] &= \begin{pmatrix} 1 & 1 & 1 & \vdots & -3 \\ 3 & 1 & -2 & \vdots & -2 \\ 2 & 4 & 7 & \vdots & 7 \end{pmatrix} \\&\approx \begin{pmatrix} 1 & 1 & 1 & \vdots & -3 \\ 0 & -2 & -5 & \vdots & 7 \\ 0 & 2 & 5 & \vdots & 13 \end{pmatrix} \begin{matrix} R_2' \rightarrow R_2 - 3R_1 \\ R_3' \rightarrow R_3 - 2R_1 \end{matrix} \\&\approx \begin{pmatrix} 1 & 1 & 1 & \vdots & -3 \\ 0 & -2 & -5 & \vdots & 7 \\ 0 & 0 & 0 & \vdots & 20 \end{pmatrix} R_3' \rightarrow R_3 + R_2\end{aligned}$$

which is the echelon form of the augmented matrix.

The reduced system is,

$$\left. \begin{aligned}x + y + z &= -3 \\-2y - 5z &= 7 \\0 &= 20\end{aligned} \right\}$$

Since, $0 = 20$ occurs, which is not possible so the given system of linear equations is inconsistent. **(Showed)**

Problem-02: Show that the following system of linear equations is consistent

$$\begin{aligned}2x - y + 3z &= 8 \\ -x + 2y + z &= 4 \\ 3x + y - 4z &= 0\end{aligned}$$

and find the solution.

Solution: The given system of linear equations is,

$$\left. \begin{aligned}2x - y + 3z &= 8 \\ -x + 2y + z &= 4 \\ 3x + y - 4z &= 0\end{aligned} \right\} \dots \dots \dots (1)$$

the system (1) can be written as,

$$AX = B \dots \dots \dots (2)$$

$$\text{where, } A = \begin{pmatrix} 2 & -1 & 3 \\ -1 & 2 & 1 \\ 3 & 1 & -4 \end{pmatrix}, X = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \text{ and } B = \begin{pmatrix} 8 \\ 4 \\ 0 \end{pmatrix}.$$

The augmented matrix is,

$$\begin{aligned}[A, B] &= \begin{pmatrix} 2 & -1 & 3 & \vdots & 8 \\ -1 & 2 & 1 & \vdots & 4 \\ 3 & 1 & -4 & \vdots & 0 \end{pmatrix} \\ &\approx \begin{pmatrix} 2 & -1 & 3 & \vdots & 8 \\ 0 & 3 & 5 & \vdots & 16 \\ 0 & 5 & -17 & \vdots & -24 \end{pmatrix} \begin{matrix} R_2' \rightarrow 2R_2 + R_1 \\ R_3' \rightarrow 2R_3 - 3R_1 \end{matrix} \\ &\approx \begin{pmatrix} 2 & -1 & 3 & \vdots & 8 \\ 0 & 3 & 5 & \vdots & 16 \\ 0 & 0 & -76 & \vdots & -152 \end{pmatrix} R_3' \rightarrow 3R_3 - 5R_2\end{aligned}$$

which is the echelon form of the augmented matrix.

The reduced system is,

$$\left. \begin{aligned}2x - y + 3z &= 8 \\ 3y + 5z &= 16 \\ -76z &= -152\end{aligned} \right\}$$

By back substitution we get, $z = 2$, $y = 2$, $x = 2$.

Hence the given system is consistent and the solution is,

$$x = 2, y = 2, z = 2.$$

Problem-03: Show that the following system of linear equations is not consistent

$$\begin{aligned}x + y + 2z + w &= 5 \\2x + 3y - z - 2w &= 2 \\4x + 5y + 3z &= 7\end{aligned}$$

Solution: The given system of linear equations is,

$$\left. \begin{aligned}x + y + 2z + w &= 5 \\2x + 3y - z - 2w &= 2 \\4x + 5y + 3z &= 7\end{aligned} \right\} \dots \dots \dots (1)$$

the system (1) can be written as,

$$AX = B \dots \dots \dots (2)$$

$$\text{where, } A = \begin{pmatrix} 1 & 1 & 2 & 1 \\ 2 & 3 & -1 & -2 \\ 4 & 5 & 3 & 0 \end{pmatrix}, X = \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} \text{ and } B = \begin{pmatrix} 5 \\ 2 \\ 7 \end{pmatrix}.$$

The augmented matrix is,

$$\begin{aligned}[A, B] &= \begin{pmatrix} 1 & 1 & 2 & 1 & \vdots & 5 \\ 2 & 3 & -1 & -2 & \vdots & 2 \\ 4 & 5 & 3 & 0 & \vdots & 7 \end{pmatrix} \\&\approx \begin{pmatrix} 1 & 1 & 2 & 1 & \vdots & 5 \\ 0 & 1 & -5 & -4 & \vdots & -8 \\ 0 & 1 & -5 & -4 & \vdots & -13 \end{pmatrix} \begin{matrix} \\ R_2' \rightarrow R_2 - 2R_1 \\ R_3' \rightarrow R_3 - 4R_1 \end{matrix} \\&\approx \begin{pmatrix} 1 & 1 & 2 & 1 & \vdots & 5 \\ 0 & 1 & -5 & -4 & \vdots & -8 \\ 0 & 0 & 0 & 0 & \vdots & -5 \end{pmatrix} \begin{matrix} \\ \\ R_3' \rightarrow R_3 - R_2 \end{matrix}\end{aligned}$$

which is the echelon form of the augmented matrix.

The reduced system is,

$$\left. \begin{aligned}x + y + 2z + w &= 5 \\y - 5z - 4w &= -8 \\0 &= -5\end{aligned} \right\}$$

Since, $0 = -5$ occurs, which is not possible so the given system of linear equations is inconsistent. **(Showned)**

Problem-04: Find for what values of λ , μ the system of linear equations

$$\begin{aligned}x + y + z &= 6 \\x + 2y + 3z &= 10 \\x + 2y + \lambda z &= \mu\end{aligned}$$

has i). a unique solution, ii). no solution, iii). infinite solutions.

Solution: The given system of linear equations is,

$$\left. \begin{aligned}x + y + z &= 6 \\x + 2y + 3z &= 10 \\x + 2y + \lambda z &= \mu\end{aligned} \right\} \dots \dots \dots (1)$$

the system (1) can be written as,

$$AX = B \dots \dots \dots (2)$$

where, $A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & \lambda \end{pmatrix}$, $X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ and $B = \begin{pmatrix} 6 \\ 10 \\ \mu \end{pmatrix}$.

The augmented matrix is,

$$\begin{aligned}[A, B] &= \begin{pmatrix} 1 & 1 & 1 & \vdots & 6 \\ 1 & 2 & 3 & \vdots & 10 \\ 1 & 2 & \lambda & \vdots & \mu \end{pmatrix} \\&\approx \begin{pmatrix} 1 & 1 & 1 & \vdots & 6 \\ 0 & 1 & 2 & \vdots & 4 \\ 0 & 1 & \lambda - 1 & \vdots & \mu - 6 \end{pmatrix} \begin{matrix} R_2' \rightarrow R_2 - R_1 \\ R_3' \rightarrow R_3 - R_1 \end{matrix} \\&\approx \begin{pmatrix} 1 & 1 & 1 & \vdots & 6 \\ 0 & 1 & 2 & \vdots & 4 \\ 0 & 0 & \lambda - 3 & \vdots & \mu - 10 \end{pmatrix} \begin{matrix} \\ R_3' \rightarrow R_3 - R_2 \end{matrix}\end{aligned}$$

which is the echelon form of the augmented matrix.

The reduced system is,

$$\left. \begin{aligned}x + y + z &= 6 \\y + 2z &= 4 \\(\lambda - 3)z &= \mu - 10\end{aligned} \right\}$$

If $\lambda \neq 3$, then the given system of equations possesses a unique solution for any value of μ .

If $\lambda = 3$ and $\mu \neq 10$, then the given system of equations possesses no solution.

If $\lambda = 3$ and $\mu = 10$, then the given system of equations possesses infinite solutions.

Problem-05: Solve the following system of linear equations

$$5x - 6y + 4z = 15$$

$$7x + 4y - 3z = 19$$

$$2x + y + 6z = 46$$

Solution: The given system of linear equations is,

$$\left. \begin{array}{l} 5x - 6y + 4z = 15 \\ 7x + 4y - 3z = 19 \\ 2x + y + 6z = 46 \end{array} \right\} \dots \dots \dots (1)$$

the system (1) can be written as,

$$AX = B \dots \dots \dots (2)$$

$$\text{where, } A = \begin{pmatrix} 5 & -6 & 4 \\ 7 & 4 & -3 \\ 2 & 1 & 6 \end{pmatrix}, X = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \text{ and } B = \begin{pmatrix} 15 \\ 19 \\ 46 \end{pmatrix}.$$

The augmented matrix is,

$$\begin{aligned} [A, B] &= \begin{pmatrix} 5 & -6 & 4 & : & 15 \\ 7 & 4 & -3 & : & 19 \\ 2 & 1 & 6 & : & 46 \end{pmatrix} \\ &\approx \begin{pmatrix} 5 & -6 & 4 & : & 15 \\ 0 & 62 & -43 & : & -10 \\ 0 & 17 & 22 & : & 200 \end{pmatrix} \begin{array}{l} R_2' \rightarrow 5R_2 - 7R_1 \\ R_3' \rightarrow 5R_3 - 2R_1 \end{array} \\ &\approx \begin{pmatrix} 5 & -6 & 4 & : & 15 \\ 0 & 62 & -43 & : & -10 \\ 0 & 0 & 2095 & : & 12570 \end{pmatrix} \begin{array}{l} \\ R_3' \rightarrow 62R_3 - 17R_2 \end{array} \end{aligned}$$

which is the echelon form of the augmented matrix.

The reduced system is,

$$\left. \begin{array}{l} 5x - 6y + 4z = 15 \\ 62y - 43z = -10 \\ 2095z = 12570 \end{array} \right\}$$

By back substitution, we get $z = 6$, $y = 4$, $x = 3$.

Hence the required result is, $x = 3$, $y = 4$, $z = 6$.

Problem-06: Solve the following system of linear equations

$$\begin{aligned} x + y - 2z + s + 3t &= 1 \\ 2x - y + 2z + 2s + 6t &= 2 \\ 3x + 2y - 4z - 3s - 9t &= 3 \end{aligned}$$

Solution: The given system of linear equations is,

$$\left. \begin{array}{l} x + y - 2z + s + 3t = 1 \\ 2x - y + 2z + 2s + 6t = 2 \\ 3x + 2y - 4z - 3s - 9t = 3 \end{array} \right\} \dots \dots \dots (1)$$

the system (1) can be written as,

$$AX = B \dots \dots \dots (2)$$

$$\text{where, } A = \begin{pmatrix} 1 & 1 & -2 & 1 & 3 \\ 2 & -1 & 2 & 2 & 6 \\ 3 & 2 & -4 & -3 & -9 \end{pmatrix}, X = \begin{pmatrix} x \\ y \\ z \\ s \\ t \end{pmatrix} \text{ and } B = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}.$$

The augmented matrix is,

$$\begin{aligned} [A, B] &= \begin{pmatrix} 1 & 1 & -2 & 1 & 3 & : & 1 \\ 2 & -1 & 2 & 2 & 6 & : & 2 \\ 3 & 2 & -4 & -3 & -9 & : & 3 \end{pmatrix} \\ &\approx \begin{pmatrix} 1 & 1 & -2 & 1 & 3 & : & 1 \\ 0 & -3 & 6 & 0 & 0 & : & 0 \\ 0 & -1 & 2 & -6 & -18 & : & 0 \end{pmatrix} \begin{array}{l} R_2' \rightarrow R_2 - 2R_1 \\ R_3' \rightarrow R_3 - 3R_1 \end{array} \\ &\approx \begin{pmatrix} 1 & 1 & -2 & 1 & 3 & : & 1 \\ 0 & -3 & 6 & 0 & 0 & : & 0 \\ 0 & 0 & 0 & -18 & -54 & : & 0 \end{pmatrix} \begin{array}{l} \\ R_3' \rightarrow 3R_3 - R_2 \end{array} \end{aligned}$$

$$\approx \begin{pmatrix} 1 & 1 & -2 & 1 & 3 & \vdots & 1 \\ 0 & 1 & -2 & 0 & 0 & \vdots & 0 \\ 0 & 0 & 0 & 1 & 3 & \vdots & 0 \end{pmatrix} \begin{matrix} R_2' \rightarrow -\frac{1}{3}R_2 \\ R_3' \rightarrow -\frac{1}{18}R_3 \end{matrix}$$

which is the echelon form of the augmented matrix.

The reduced system is,

$$\begin{cases} x + y - 2z + s + 3t = 1 \\ y - 2z = 0 \\ s + 3t = 0 \end{cases}$$

There are 3 equations in 5 unknowns, so there are $(5-3)=2$ free variables which are z and t . Thus the system is consistent with an infinite number of solutions.

We put $z = a$ and $t = b$. So by back substitution we have, $s = -3b$, $y = 2a$, $x = 1$.

Hence the required result is, $x = 1$, $y = 2a$, $z = a$, $s = -3b$, $t = b$.

Problem-07: Solve the following system of linear equations

$$\begin{aligned} 2x + 3y + 5z + t &= 3 \\ 3x + 4y + 2z + 3t &= -2 \\ x + 2y + 8z - t &= 8 \\ 7x + 9y + z + 8t &= 0 \end{aligned}$$

Solution: The given system of linear equations is,

$$\begin{cases} 2x + 3y + 5z + t = 3 \\ 3x + 4y + 2z + 3t = -2 \\ x + 2y + 8z - t = 8 \\ 7x + 9y + z + 8t = 0 \end{cases} \dots \dots \dots (1)$$

the system (1) can be written as,

$$AX = B \dots \dots \dots (2)$$

$$\text{where, } A = \begin{pmatrix} 2 & 3 & 5 & 1 \\ 3 & 4 & 2 & 3 \\ 1 & 2 & 8 & -1 \\ 7 & 9 & 1 & 8 \end{pmatrix}, X = \begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix} \text{ and } B = \begin{pmatrix} 3 \\ -2 \\ 8 \\ 0 \end{pmatrix}.$$

The augmented matrix is,

$$\begin{aligned}
[A, B] &= \begin{pmatrix} 2 & 3 & 5 & 1 & : & 3 \\ 3 & 4 & 2 & 3 & : & -2 \\ 1 & 2 & 8 & -1 & : & 8 \\ 7 & 9 & 1 & 8 & : & 0 \end{pmatrix} \\
&\approx \begin{pmatrix} 1 & 2 & 8 & -1 & : & 8 \\ 3 & 4 & 2 & 3 & : & -2 \\ 2 & 3 & 5 & 1 & : & 3 \\ 7 & 9 & 1 & 8 & : & 0 \end{pmatrix} R_1 \leftrightarrow R_3 \\
&\approx \begin{pmatrix} 1 & 2 & 8 & -1 & : & 8 \\ 0 & -2 & -22 & 6 & : & -26 \\ 0 & -1 & -11 & 3 & : & -13 \\ 0 & -5 & -55 & 15 & : & -56 \end{pmatrix} \begin{matrix} R_2' \rightarrow R_2 - 3R_1 \\ R_3' \rightarrow R_3 - 2R_1 \\ R_4' \rightarrow R_4 - 7R_1 \end{matrix} \\
&\approx \begin{pmatrix} 1 & 2 & 8 & -1 & : & 8 \\ 0 & -2 & -22 & 6 & : & -26 \\ 0 & 0 & 0 & 0 & : & 0 \\ 0 & 0 & 0 & 0 & : & 18 \end{pmatrix} \begin{matrix} R_3' \rightarrow 2R_3 - R_2 \\ R_4' \rightarrow 2R_4 - 5R_2 \end{matrix} \\
&\approx \begin{pmatrix} 1 & 2 & 8 & -1 & : & 8 \\ 0 & 1 & 11 & -3 & : & 13 \\ 0 & 0 & 0 & 0 & : & 0 \\ 0 & 0 & 0 & 0 & : & 18 \end{pmatrix} R_2' \rightarrow -\frac{1}{2}R_2
\end{aligned}$$

which is the echelon form of the augmented matrix.

The reduced system is,

$$\begin{aligned}
&\left. \begin{aligned} x+2y+8z-t &= 8 \\ y+11z-3t &= 13 \\ 0 &= 0 \\ 0 &= 18 \end{aligned} \right\} \\
\text{or, } &\left. \begin{aligned} x+2y+8z-t &= 8 \\ y+11z-3t &= 13 \\ 0 &= 18 \end{aligned} \right\}
\end{aligned}$$

Since, $0=18$ occurs, which is not possible so the given system of linear equations is inconsistent and it has no solution.

Problem-08: Solve the following system of linear equations

$$\begin{aligned}
x+2y-2z-t &= 0 \\
2x+5y-3z-t &= 1 \\
3x+8y-4z-t &= 2 \\
x+5y+z+2t &= 3
\end{aligned}$$

Solution: The given system of linear equations is,

$$\left. \begin{aligned}
x+2y-2z-t &= 0 \\
2x+5y-3z-t &= 1 \\
3x+8y-4z-t &= 2 \\
x+5y+z+2t &= 3
\end{aligned} \right\} \dots \dots \dots (1)$$

the system (1) can be written as,

$$AX = B \dots \dots \dots (2)$$

$$\text{where, } A = \begin{pmatrix} 1 & 2 & -2 & -1 \\ 2 & 5 & -3 & -1 \\ 3 & 8 & -4 & -1 \\ 1 & 5 & 1 & 2 \end{pmatrix}, X = \begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix} \text{ and } B = \begin{pmatrix} 0 \\ 1 \\ 2 \\ 3 \end{pmatrix}.$$

The augmented matrix is,

$$\begin{aligned}
[A, B] &= \begin{pmatrix} 1 & 2 & -2 & -1 & \vdots & 0 \\ 2 & 5 & -3 & -1 & \vdots & 1 \\ 3 & 8 & -4 & -1 & \vdots & 2 \\ 1 & 5 & 1 & 2 & \vdots & 3 \end{pmatrix} \\
&\approx \begin{pmatrix} 1 & 2 & -2 & -1 & \vdots & 0 \\ 0 & 1 & 1 & 1 & \vdots & 1 \\ 0 & 2 & 2 & 2 & \vdots & 2 \\ 0 & 3 & 3 & 3 & \vdots & 3 \end{pmatrix} \begin{array}{l} R_2' \rightarrow R_2 - 2R_1 \\ R_3' \rightarrow R_3 - 3R_1 \\ R_4' \rightarrow R_4 - R_1 \end{array} \\
&\approx \begin{pmatrix} 1 & 2 & -2 & -1 & \vdots & 0 \\ 0 & 1 & 1 & 1 & \vdots & 1 \\ 0 & 0 & 0 & 0 & \vdots & 0 \\ 0 & 0 & 0 & 0 & \vdots & 0 \end{pmatrix} \begin{array}{l} R_3' \rightarrow R_3 - 2R_1 \\ R_4' \rightarrow R_4 - 3R_2 \end{array}
\end{aligned}$$

which is the echelon form of the augmented matrix.

The reduced system is,

$$\left. \begin{array}{l} x+2y-2z-t=0 \\ y+z+t=1 \\ 0=0 \\ 0=0 \end{array} \right\}$$

$$\text{or, } \left. \begin{array}{l} x+2y-2z-t=0 \\ y+z+t=1 \end{array} \right\}$$

There are 2 equations in 4 unknowns, so there are $(4-2) = 2$ free variables which are z and t . Thus the system is consistent with an infinite number of solutions.

We put $z = a$ and $t = b$. So by back substitution we have, $y = 1 - a - b$, $x = 4a + 3b - 2$.

Hence the required result is, $x = 4a + 3b - 2$, $y = 1 - a - b$, $z = a$, $t = b$.

Exercise:

1. Solve the following system of linear equations

$$3x + 5y - 7z = 13$$

$$4x + y - 12z = 6$$

$$2x + 9y - 3z = 20$$

2. Solve the following system of linear equations

$$x + y + z = 3$$

$$x + 2y + 3z = 4$$

$$x + 4y + 9z = 6$$

3. Solve the following system of linear equations

$$2x - 3y + 4z = 1$$

$$3x + 4y - 5z = 1$$

$$5x - 7y + 2z = 3$$

4. Determine for what value of a the following system of linear equations has

$$x + y - z = 1$$

$$2x + 3y + az = 3$$

$$x + ay + 3z = 2$$

i). no solution, ii). more than one solution, iii). a unique solution.

5. Determine for what values of λ and μ the following system of linear equations has

$$2x + 3y + z = 5$$

$$3x - y + \lambda z = 2$$

$$x + 7y - 6z = \mu$$

i). no solution, ii). more than one solution, iii). a unique solution.

6. Solve the following system of linear equations

$$x + y + z + t = 4$$

$$x + 2y + 2z - t = 4$$

$$x + 4y + 9z + 2t = 16$$